Low-Latency Deep Learning-Based Denoising of Self-Generated Footstep Noise in Quadruped Robots for Remote Acoustic Situational Awareness

Hyeon Jae Shin[1]; Jin Tae Seok[2]; So Yeong Park[3]; Seon Man Kim[*]
School of Computing and Artificial Intelligence, Hanshin University, South Korea

ABSTRACT

In disaster response scenarios, quadruped robots are increasingly deployed to support remote situational awareness by transmitting real-time audiovisual data from hazardous environments. However, their onboard microphones often capture high-amplitude, periodic footstep noise generated by the robot itself, which severely degrades the intelligibility of critical ambient sounds. To address this, this research focuses on developing a low-latency, deep learning-based denoising algorithm suitable for on-robot deployment. The proposed method modifies the Fair-Denoiser architecture, a powerful U-Net based model, to optimize for real-time performance and computational efficiency. We introduce targeted architectural changes and a novel loss function to enhance performance. The model was trained and evaluated on the Valentini benchmark dataset to establish its general-purpose denoising capabilities under controlled conditions. Evaluation results demonstrate that our proposed model surpasses strong baselines like the original Fair-Denoiser and CleanUNet, achieving a PESQ score of over 3.0 and a STOI score of over 0.94. This work validates the model's suitability for real-time applications, establishing a critical foundation for future deployment and fine-tuning on specialized robotic noise datasets.

Keywords: Denoising, Quadruped Robot, Robot Noise

1. INTRODUCTION

The deployment of quadruped robots in disaster response and hazardous environment monitoring has marked a significant advancement in remote situational awareness. These platforms can navigate complex and dangerous terrains inaccessible to human first responders, providing critical audiovisual data to remote operators. While visual information is essential, real-time audio transmission is equally paramount, as it can carry determinative cues such as the voices of trapped survivors, calls for assistance, or the sounds of shifting debris indicating structural instability. The clarity of this acoustic channel can directly influence the speed and efficacy of rescue operations.

However, a fundamental challenge arises from the robot's own operation. The mechanical action of the robot, particularly the high-amplitude, periodic impact of its footsteps and the whine of its motors, generates significant "ego-noise." This self-generated noise is captured by the onboard microphones and can easily mask the subtle, yet vital, ambient sounds that are the objective of the acoustic monitoring mission. The structured and impulsive nature of this footstep noise presents a unique and difficult challenge for traditional denoising algorithms.

The ultimate goal of this research is to develop a comprehensive acoustic situational awareness system that can intelligently separate critical environmental sounds from the robot's own ego-noise, as illustrated in Figure 1. In a typical disaster scenario, the raw audio captured by the robot is a complex mixture of target sounds (e.g., emergency speech) and noise (e.g., robot footsteps and motor whine). To provide a clear audio feed to the remote operator, a multi-stage process is envisioned: first, the dominant, structured ego-noise from footsteps and motors must be removed. Following this, a sound source separation module can distinguish between enhanced speech and other crucial environmental sounds. This paper focuses on the first and most

[1] sjt2765@hs.ac.kr
[2] guswotls123@gmail.com
[3] kong02931@gmail.com
* kobem30002@gmail.com

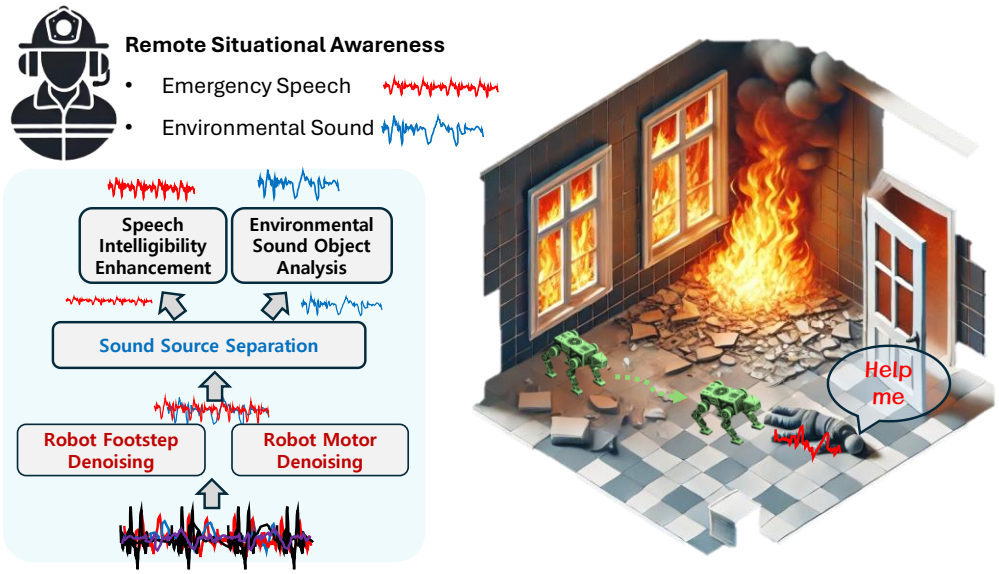


Figure 1 – The overall concept for achieving remote acoustic situational awareness. The system first denoises the robot's self-generated noise (footsteps, motor) from the raw audio signal before separating critical sounds like emergency speech.

critical stage of this pipeline: the effective and real-time removal of robot footstep noise. Addressing this primary source of contamination is a necessary prerequisite for any subsequent analysis and is therefore the central justification for the present work.

For any on-robot processing system intended to support a human operator, latency is a non-negotiable constraint. To be effective in a dynamic disaster scenario, the audio denoising process must occur in real-time. This project operates under a strict processing delay requirement, rendering many computationally intensive, state-of-the-art denoising models unsuitable for deployment. This necessitates a focused effort on developing lightweight and computationally efficient model architectures. While the ultimate goal is to suppress robotic noise, the initial phase of this research focuses on creating a model with superior denoising performance and real-time capabilities, validated on a standardized public dataset. A specialized dataset of robotic footstep noise is currently under construction and was not used in this phase.

In recent years, deep learning-based speech enhancement has been dominated by end-to-end models that operate directly on the raw audio waveform [1]. Architectures based on the U-Net structure, such as Fair-Denoiser [2] and CleanUNet [3], have demonstrated remarkable performance. This paper presents a novel denoising model specifically tailored for real-time performance. Our primary contribution is a modified U-Net architecture, derived from the Fair-Denoiser model, which has been systematically optimized for both performance and computational efficiency. We detail architectural changes and introduce a novel loss function, demonstrating superior performance against established baselines on the Valentini benchmark dataset [4].

2. The Fair-Denoiser Base Architecture

The foundation of our proposed model is the Fair-Denoiser, a causal, real-time speech enhancement model that operates directly on the raw audio waveform [2]. This model is an adaptation of the DEMUCS architecture, a U-Net-like convolutional structure originally developed for music source separation, and its end-to-end, waveform-to-waveform approach makes it well-suited for high-fidelity audio processing tasks [5].

The model is structured as a fully convolutional encoder-decoder network featuring a recurrent bottleneck and U-Net-style skip connections [2]. The process begins with the encoder network, which is composed of a series of L layers designed to progressively downsample the input waveform and extract hierarchical feature representations. Each encoder layer consists of a strided one-dimensional convolution (Conv1D) with kernel size K and stride S, followed by a Rectified Linear Unit (ReLU) activation. This is then passed through a 1x1 convolution and a Gated Linear Unit (GLU), which acts as a learned gating mechanism to selectively pass or block information. At the deepest level of the architecture, the compressed feature representation is processed by a 2-layer Long Short-Term Memory (LSTM) network, which serves as the model's bottleneck [2]. The

Table 1 – Architectural comparison of base and modified models

Parameter	Base Model (Fair-Denoiser)	Proposed Model (Mod 1)	Lightweight Alternative (Mod 2)
Layer Depth	8	8	5
CNN Bias	Local	Global	Global
CNN Kernel Size & Stride	8 & 4	8 & 4	4 & 2
Encoder Activation	GLU	GLU	GLU
Decoder Activation	GLU	ReLU	ReLU
Max Channels	768	768	576
Resampling	Enabled	Enabled	Disabled
Loss Function	Multi-resolution STFT	Multi-dynamic range & Multi-res STFT	Multi-dynamic range & Multi-res STFT

LSTM is critical for capturing the long-range temporal dependencies and contextual information within the audio signal. The decoder network then mirrors the encoder's structure, tasked with upsampling the feature representation to reconstruct the clean audio waveform using transposed convolutions to increase the temporal resolution. To ensure a high-fidelity output, skip connections—a hallmark of the U-Net architecture—link the output of each encoder layer directly to the input of its corresponding decoder layer [5]. This is crucial for preserving fine details in the audio waveform and preventing the information loss that can occur in deep networks.

The original Fair-Denoiser model is trained using a composite loss function that operates in both the time and frequency domains. The total loss is a weighted sum of an L1 loss on the raw waveform and a multi-resolution Short-Time Fourier Transform (STFT) loss [2]. This hybrid approach ensures that the model not only matches the waveform sample-by-sample but also preserves the spectral structure of the speech, which is highly correlated with human perception of audio quality.

3. Proposed Modified Denoiser for Real-Time Performance

The primary design objective was to enhance denoising performance and efficiency for real-time applications. This goal guided a series of deliberate architectural changes to the Fair-Denoiser baseline, as summarized in Table 1. Our final proposed model, "Modification 1," focuses on improving performance through targeted changes that do not alter the core network complexity. For applications where latency and model size are the absolute highest priorities, we also present "Modification 2" as a more aggressive optimization.

3.1 Proposed Model (Modification 1)

Our final proposed model focuses on improving performance and efficiency through targeted changes that do not alter the core complexity of the network. One such change is the replacement of the Gated Linear Unit (GLU) with the standard Rectified Linear Unit (ReLU) in the decoder layers. While GLUs offer a more powerful, learned gating mechanism, this capability comes at a computational cost; by switching to the faster ReLU activation, we gain a tangible increase in execution speed.

A second key innovation is the introduction of the "Multi-dynamic range & Multi-resolution STFT" (MDRR-STFT) loss function. This loss addresses a key limitation of standard STFT-based losses, which tend to be dominated by high-energy spectral components, causing the model to neglect quieter but perceptually important details like unvoiced consonants. The MDRR-STFT loss mitigates this by partitioning the spectrogram of each frame into multiple dynamic range bands (e.g., high, mid, low energy) based on their relative decibel levels. The loss is then calculated independently for each band i as a sum of spectral convergence (L_{sc}) and log-magnitude (L_{mag}) losses :

$$L_{\text{STFT}}^{(i)} = L_{\text{sc}}^{(i)} + L_{\text{mag}}^{(i)} \tag{1}$$

By computing the loss across these separate energy bands, the model is forced to learn from both high-energy and low-energy speech components, preventing weaker signals from being overlooked during optimization and leading to improved overall intelligibility and preservation of fine spectral details.

Table 2 – Denoising performance comparison on the Valentini test set

Model	Parameters, M	PESQ (WB)	STOI,%
Noisy	-	1.94	92.7
FAIR-denoiser	73.70	2.93	94.7
CleanUNet	180.0	2.99	94.9
Proposed Model (Mod 1)	73.70	3.04	94.9
Proposed Model (Mod 2)	**31.0**	2.89	94.6

3.2 Lightweight Alternative (Modification 2)

For applications where latency and model size are the absolute highest priorities, we propose a second, more lightweight model that incorporates all the changes from Modification 1 and adds further complexity reductions. To minimize the computational load, we reduced the number of encoder-decoder layers from 8 to 5, the maximum channel count from 768 to 576, and the convolutional kernel size from 8 to 4, with the stride correspondingly halved from 4 to 2. These changes collectively yield a substantial reduction in the model's total parameter count and computational demands. Furthermore, the computationally intensive pre/post-processing resampling step is disabled, eliminating this overhead entirely and further reducing end-to-end latency.

4. EXPERIMENTAL SETUP

To validate our proposed models, we conducted a comprehensive experiment on the Valentini dataset from the Voice Bank corpus, a benchmark designed to evaluate performance in domain-mismatched scenarios [4]. All audio, originally sampled at 48 kHz, was resampled to 16 kHz. The training set utilized 28 speakers and consisted of 11,572 clean/noisy pairs mixed with 10 different noise types at SNR levels of 0, 5, 10, and 15 dB [4]. The test set, ensuring variability, used 2 different speakers and comprised 824 pairs mixed with 5 distinct noise types at SNRs of 2.5, 7.5, 12.5, and 17.5 dB. For training, 4.5-second clips were randomly extracted and augmented using Remix, Band-mask, and Revecho techniques. The models were trained for a maximum of 400 epochs with a batch size of 64 and an early-stopping parameter of 300, using the Adam optimizer (β_1 =0.9, β_2 ==0.999) on six NVIDIA A40 GPUs. A linear warmup with a cosine annealing learning rate schedule was employed, with a maximum learning rate of 2×10−4 and a 5% warmup ratio, optimizing a full-band short-term Fourier transform loss. Performance was quantitatively assessed using two industry-standard metrics: Perceptual Evaluation of Speech Quality (PESQ) to measure perceptual quality and Short-Time Objective Intelligibility (STOI) to measure speech intelligibility [6].

5. RESULTS AND DISCUSSION

The effectiveness of our proposed modifications was evaluated against the base Fair-Denoiser and the state-of-the-art CleanUNet model, with the results summarized in Table 2 demonstrating the success of our approach. Our final proposed model, "Modification 1," achieved the highest performance across all tested configurations, delivering a PESQ score of 3.03537 and a STOI of 0.94891, surpassing both the original Fair-Denoiser and the strong CleanUNet baseline. This result empirically validates the effectiveness of our architectural changes, particularly the new multi-dynamic range loss function and the switch to ReLU in the decoder, which together provide superior denoising quality without increasing model complexity.

In contrast, the "Lightweight Alternative (Modification 2)" underwent aggressive complexity reductions, leading to an expected slight decrease in the PESQ score to 2.88561. Crucially, however, its STOI score remained exceptionally high at 0.94606, indicating almost no degradation in speech intelligibility. This outcome represents a successful and intentional engineering trade-off, as the substantial reduction in architectural complexity is precisely what enables the model to achieve minimal latency, making it a compelling option for deployment on hardware with extreme resource constraints. Therefore, the choice between Modification 1 and 2 depends on the specific application's priority: maximum perceptual quality versus minimum latency.

6. CONCLUSIONS

This paper presented the design and evaluation of a deep learning-based denoising algorithm optimized for real-time performance, with the ultimate goal of deployment on resource-constrained robotic platforms. Through targeted modifications to the Fair-Denoiser architecture, specifically by introducing a novel multi-dynamic range loss function and simplifying decoder activations, we developed a model that outperforms strong baselines on the standardized Valentini benchmark, achieving a superior balance of performance and efficiency with high PESQ and STOI scores. Additionally, we presented a lightweight alternative model that prioritizes minimal latency and model size through significant complexity reductions. While this alternative exhibits a minor drop in perceptual quality, it maintains excellent speech intelligibility, making it a viable candidate for the most demanding real-time applications. This research successfully validates our architectural modifications and establishes a robust, high-performing, and efficient denoising model. The technology developed provides a critical foundation for enhancing the acoustic situational awareness of robots, and future work will focus on deploying this model onto a physical quadruped robot and fine-tuning it using a custom-built dataset of robotic ego-noise to transition from a general-purpose denoiser to a specialized system for real-world operational environments.

ACKNOWLEDGEMENTS

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (NRF-2022R1A2C2010614).

REFERENCES

1. Benesty J, Makino S, Chen J. Speech Enhancement. New York, USA: Springer; 2005.
2. Défossez A, Synnaeve G, Adi Y. Real Time Speech Enhancement in the Waveform Domain. Proc Interspeech 2020; 25-29 October 2020; Shanghai, China. 2020. p. 3291-5.
3. Kong Z, Ping W, Dantrey A, Catanzaro B. Speech Denoising in the Waveform Domain with Self-Attention. Proc ICASSP 2022; 22-27 May 2022; Singapore. 2022. p. 7772-6.
4. Stoller D, Ewert S, Dixon S. Wave-U-Net: A Multi-Scale Neural Network for End-to-End Audio Source Separation. Proc 19th International Society for Music Information Retrieval Conference; 23-27 September 2018; Paris, France. 2018. p. 334-40.
5. Valentini-Botinhao C, Wang X, Takaki S, Yamagishi J. Speech Enhancement for a Noise-Robust Text-to-Speech Synthesis System Using Deep Recurrent Neural Networks. Proc Interspeech 2016; 8-12 September 2016; San Francisco, USA. 2016. p. 352-6.
6. Taal CH, Hendriks RC, Heusdens R, Jensen J. An algorithm for intelligibility prediction of time-frequency weighted noisy speech. IEEE Transactions on Audio, Speech, and Language Processing. 2011;19(7):2125-36.